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CO3-AT2-MCQ

Total points **29/32** ?

CSA1223

✗ Registration Number *

192473008

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✓ 19. A pipeline hazard is a situation that: *

- ☐ A) Increases memory capacity
- ☒ B) Prevents the next instruction from executing normally
- ☐ C) Reduces instruction size
- ☐ D) Removes the need for registers

✓

✗ Name of the Student *

.../1

Poorvaja B

✗

29/30
97%
1/1

✓ 10. Multiple-bus organization improves processor performance mainly by: *1/1

- ☐ Increasing instruction length
- ☒ Allowing simultaneous movement of data ✓
- ☐ Decreasing register capacity
- ☐ Removing the need for control signals

✗ 20. Which of the following is a data hazard? * 0/1

- ☒ A) Structural hazard ✗
- ☐ B) RAW dependency
- ☐ C) Branch prediction
- ☐ D) Cache hit

Correct answer

- ☒ B) RAW dependency

✓ 15. In a typical five-stage instruction pipeline, which stage is responsible for performing arithmetic or logical operations? *1/1

- ☐ IF
- ☐ ID
- ☒ EX ✓
- ☐ WB



28. The dispatch unit in a superscalar processor is responsible for: * 1/1

- ☐ A) Storing data permanently
- ☒ B) Sending instructions to appropriate execution units
- ☐ C) Generating external clock signals
- ☐ D) Managing input/output devices



27. Speculative execution refers to: * 1/1

- ☒ A) Executing instructions based on predicted future control flow
- ☐ B) Executing only completed instructions
- ☐ C) Executing instructions only after program termination
- ☐ D) Avoiding all branch instructions



✓ 29. Simultaneous Multithreading (SMT) allows a processor core to: * 1/1

- ☒ A) Execute instructions from multiple threads concurrently
- ☐ B) Increase the physical number of cores automatically
- ☐ C) Eliminate pipeline stages
- ☐ D) Use only one execution unit



✓ 14. Which of the following is a typical stage of an instruction pipeline? * 1/1

☒ Instruction Fetch ✓

☐ Data Encryption

☐ Program Compilation

☐ File Storage

✓ 1. Which notation is commonly used to represent a transfer of data from register R1 to register R2? *1/1

☐ $R1 + R2$

☒ $R1 \rightarrow R2$ ✓

☐ $R1 \times R2$

☐ $R1 - R2$

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✓ 6. Which operation is generally used to set selected bits of a register to 1? *1/1

☐ AND

☒ OR ✓

☐ NOT

☐ XOR

✓ 30. Intel's Hyper-Threading technology is an implementation of. * 1/1

☐ A) Superscalar pipelining only

☒ B) Simultaneous Multithreading (SMT) ✓

☐ C) Cache mapping

☐ D) Virtual memory

✓ 17. Pipeline performance is primarily measured using: * 1/1

☒ A) Instruction throughput and execution time ✓

☐ B) Register size only

☐ C) Memory capacity only

☐ D) Number of instructions in the program only

✓ 16. The correct sequence of a typical five-stage pipeline is: *

1/1

- ☒ IF → ID → EX → MEM → WB ✓
- ☐ ID → IF → MEM → EX → WB
- ☐ EX → IF → ID → WB → MEM
- ☐ IF → EX → ID → MEM → WB

✓ 18. Ideally, after a pipeline is filled, one instruction can be completed: * 1/1

- ☒ A) Every clock cycle ✓
- ☐ B) Every five clock cycles
- ☐ C) Only after all instructions finish
- ☐ D) Every two clock cycles

✓ 13. The main purpose of instruction pipelining is to: *

1/1

- ☐ Reduce instruction set size
- ☒ Increase instruction throughput ✓
- ☐ Eliminate registers
- ☐ Reduce memory capacity

✓ 5. The operation $R3 \leftarrow R1 + R2$ performs. *

1/1

- ☐ Logical operation
- ☒ Arithmetic operation
- ☐ Shift operation
- ☐ Rotate operation

✓

✓ 9. In a three-bus organization, the ALU can typically receive operands from: *1/1

- ☐ Only memory
- ☒ Two separate buses
- ☐ One bus only
- ☐ The control unit only

✓

✓ 4. Which of the following is a logical micro-operation? *

1/1

- ☐ Addition
- ☐ Subtraction
- ☐ Increment
- ☒ AND

✓

✓ 26. The primary benefit of out-of-order execution is. *

1/1

- ☒ A) Better utilization of processor resources ✓
- ☐ B) Reduced register count
- ☐ C) Smaller memory size
- ☐ D) Elimination of branch instructions

✓ 3. Which operation is an arithmetic micro-operation? *

1/1

- ☐ AND
- ☐ OR
- ☒ Addition ✓
- ☐ XOR

✓ 25. Out-of-order execution allows a processor to: *

1/1

- ☐ A) Execute instructions only in program order
- ☒ B) Execute independent instructions before earlier stalled instructions ✓
- ☐ C) Eliminate instruction decoding
- ☐ D) Disable pipelining

✓ 12. Consider the operation $R3 \leftarrow R1 + R2$. In a two-input-bus organization, R1 and R2 can be placed on:

*1/1

- ☐ The same bus only
- ☒ Two different input buses ✓
- ☐ The memory bus only
- ☐ The control bus

✓ 8. The primary advantage of a multiple-bus organization is: *

1/1

- ☐ Reduced memory size
- ☒ Increased parallel data transfers ✓
- ☐ Reduced number of registers
- ☐ Elimination of the ALU

✓ 22. Which hazard is directly associated with branch instructions? *

1/1

- ☒ A) Control hazard ✓
- ☐ B) Data hazard
- ☐ C) Structural hazard
- ☐ D) Memory hazard

✓ 11. In a multiple-bus system, buses are primarily used to: *

1/1

- ☐ Store instructions permanently
- ☒ Transfer data between processor components ✓
- ☐ Decode instructions
- ☐ Generate clock signals

✓ 24. A superscalar processor is capable of: *

1/1

- ☐ A) Executing only one instruction per clock cycle
- ☒ B) Issuing multiple instructions in a single clock cycle ✓
- ☐ C) Executing instructions without registers
- ☐ D) Eliminating the ALU

✓ 21. A Read After Write (RAW) hazard occurs when: *

1/1

- ☒ A) An instruction reads a value before a previous instruction writes it ✓
- ☐ B) Two instructions write simultaneously
- ☐ C) An instruction writes before reading
- ☐ D) Two instructions fetch simultaneously

✓ 23. A structural hazard occurs when: *

1/1

- ☒ A) Two instructions require the same hardware resource simultaneously ✓
- ☐ B) An instruction depends on another instruction's result
- ☐ C) A branch is incorrectly predicted
- ☐ D) Data is stored in a register

✓ 7. The XOR operation produces a 1 when: *

1/1

- ☐ Both inputs are 0
- ☐ Both inputs are 1
- ☒ Inputs are different ✓
- ☐ Inputs are always equal

✓ 2. In register transfer language, the statement $R2 \leftarrow R1$ indicates that: * 1/1

- ☐ Contents of R2 are transferred to R1
- ☒ Contents of R1 are copied into R2 ✓
- ☐ R1 and R2 are added
- ☐ R1 is cleared